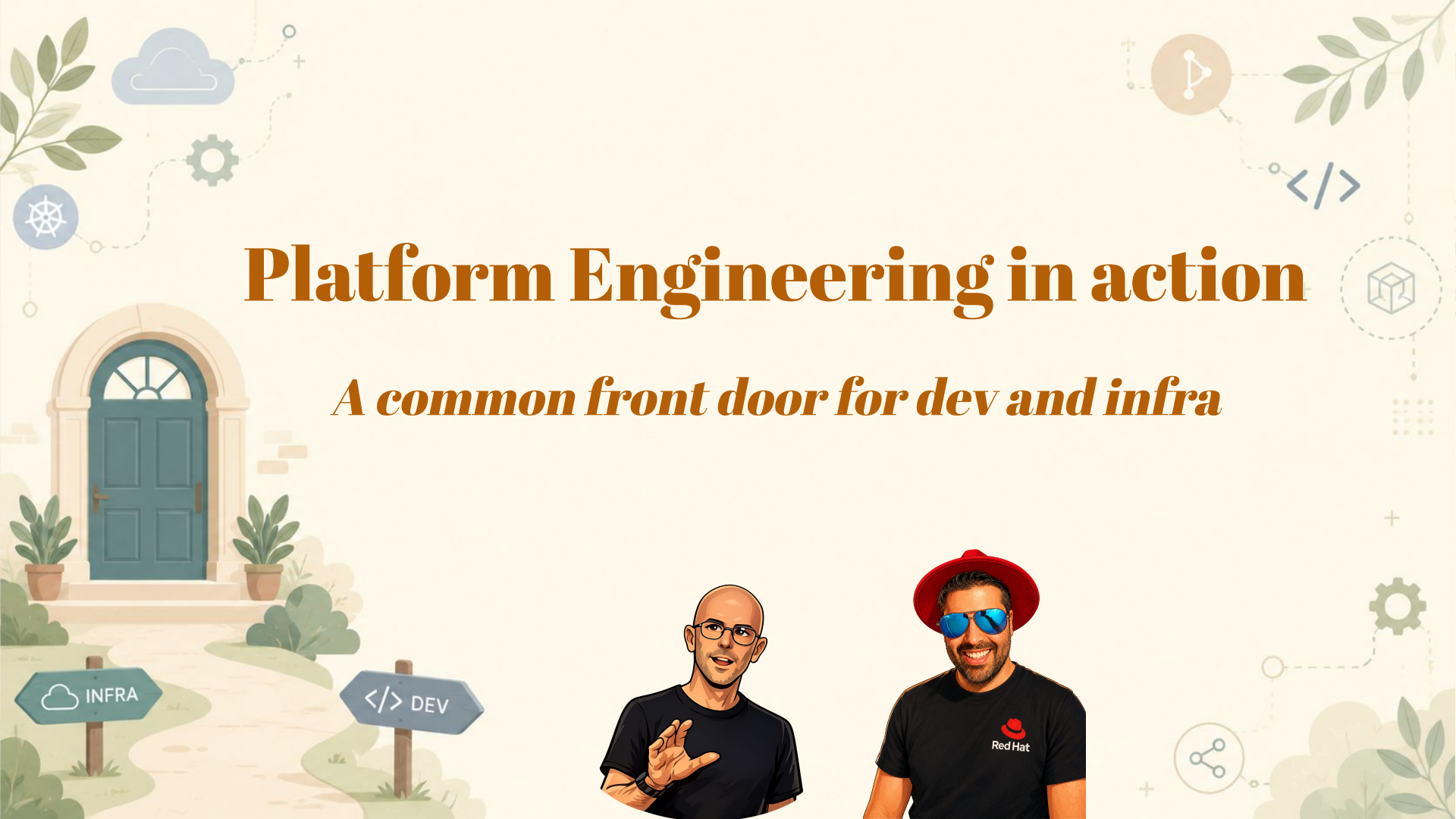


# Platform Engineering in action

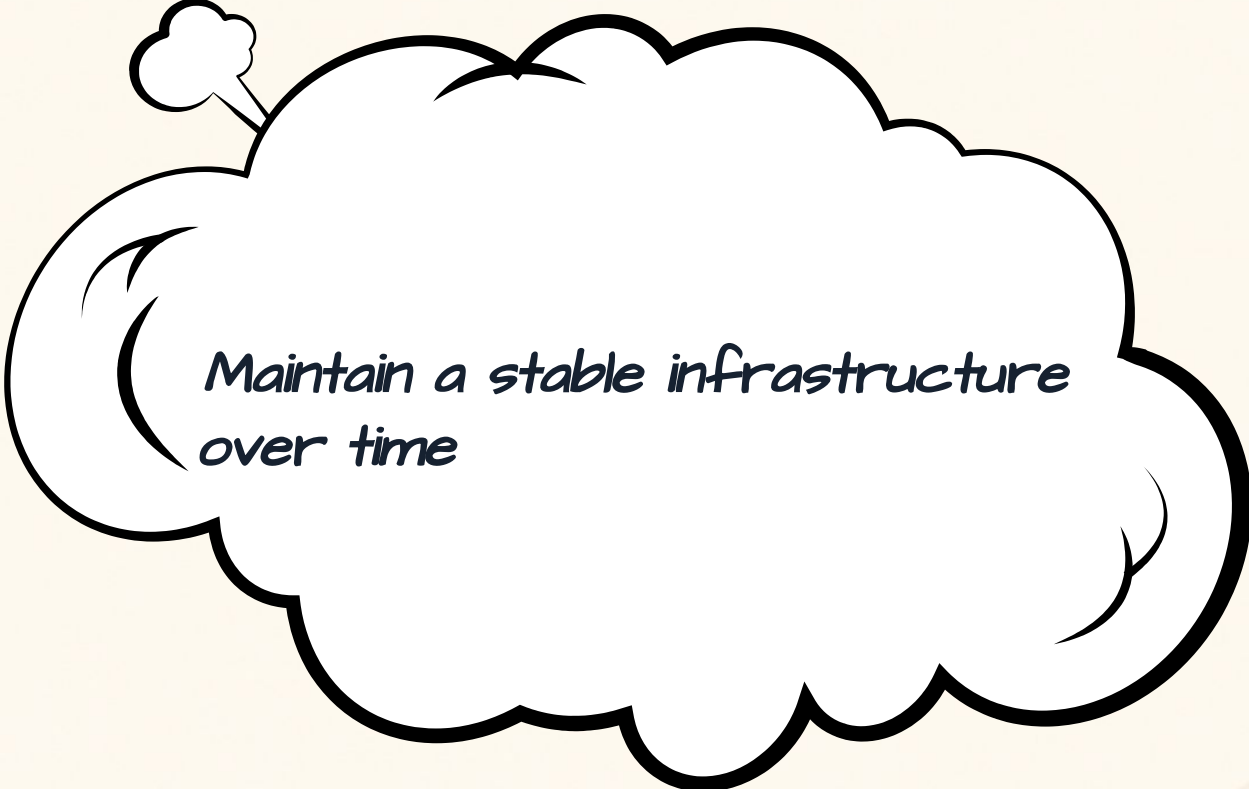
*A common front door for dev and infra*





- Day in the live of infra
- Day in the live of dev
- Platform engineering
- Dev and infra come together
- Demo





*Maintain a stable infrastructure  
over time*

Non-deterministic



Non-deterministic

Non-deterministic



Automation



Deterministic

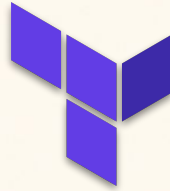


Ansible

How does infra automation  
look like?



OpenTofu



Terraform

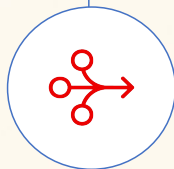


Shell

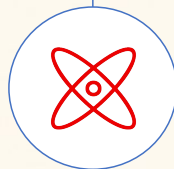
# The infra control center

Screenshot of a code editor showing an Ansible playbook named 'playbook.yml'. The playbook contains tasks for updating web servers, writing an Apache config file, updating database servers, and ensuring PostgreSQL is started. A chat window on the right shows a conversation with 'CLAUDE CODE'.

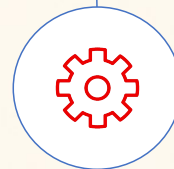
```
1 ---
2 - name: Update web servers
3   hosts: webserver
4   become: true
5
6
7   tasks:
8     - name: Ensure apache is at the present version
9       ansible.builtin.dnf:
10         name: httpd
11         state: present
12
13     - name: Write the apache config file
14       ansible.builtin.template:
15         src: /srv/httpd.j2
16         dest: /etc/httpd.conf
17         mode: "0644"
18
19 - name: Update db servers
20   hosts: databases
21   become: true
22
23   tasks:
24     - name: Ensure postgresql is at the present version
25       ansible.builtin.dnf:
26         name: postgresql
27         state: present
28
29     - name: Ensure that postgresql is started
30       ansible.builtin.service:
31         name: postgresql
32         state: started
```



**Simple language**  
And pretty readable



**Powerful**  
Large ecosystem  
Collections



**Agentless**  
Easy integration in  
your environments

```
---
- name: Install and configure Apache HTTP Server
  hosts: all
  become: yes

  tasks:
    - name: Install httpd package
      ansible.builtin.dnf:
        name: httpd
        state: present

    - name: Ensure httpd is running and enabled
      ansible.builtin.service:
        name: httpd
        state: started
        enabled: yes
```

The background is a light cream color with several decorative elements. In the top-left corner, there is a light orange abstract shape containing a small orange circle, an orange plus sign, and a green plus sign. In the top-right corner, there is a 3x3 grid of small green dots. On the left side, there are two horizontal wavy lines. On the right side, there are two horizontal wavy lines. In the bottom-left corner, there is a 3x3 grid of small green dots. In the bottom-right corner, there is a light orange abstract shape containing a small green circle and an orange plus sign.

Fire and forget?



# Day in the life of a developer



# As a developer...

Too much time lost in chaos



My main challenges...

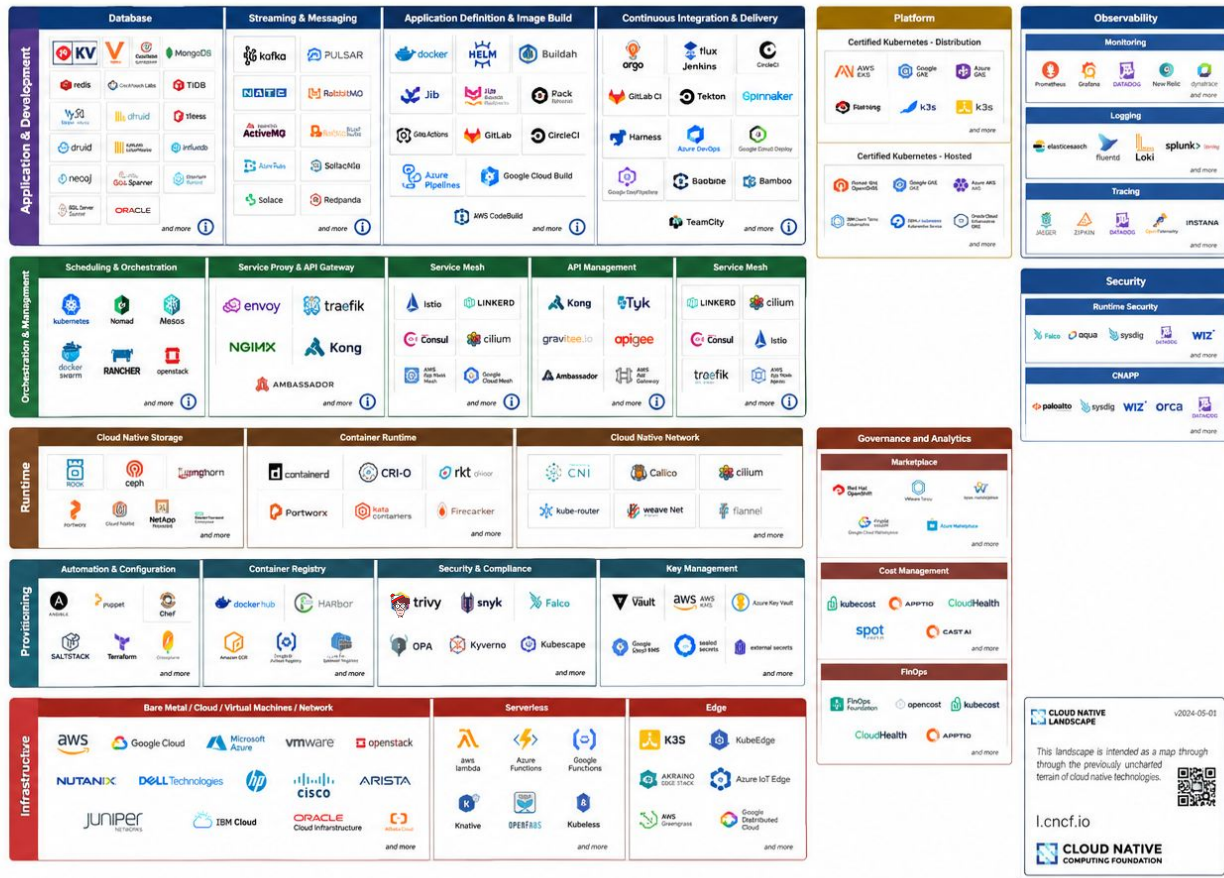


# The Cloud-Native Grocery Store

Where's Waldo?

or Wally?

or Charly?

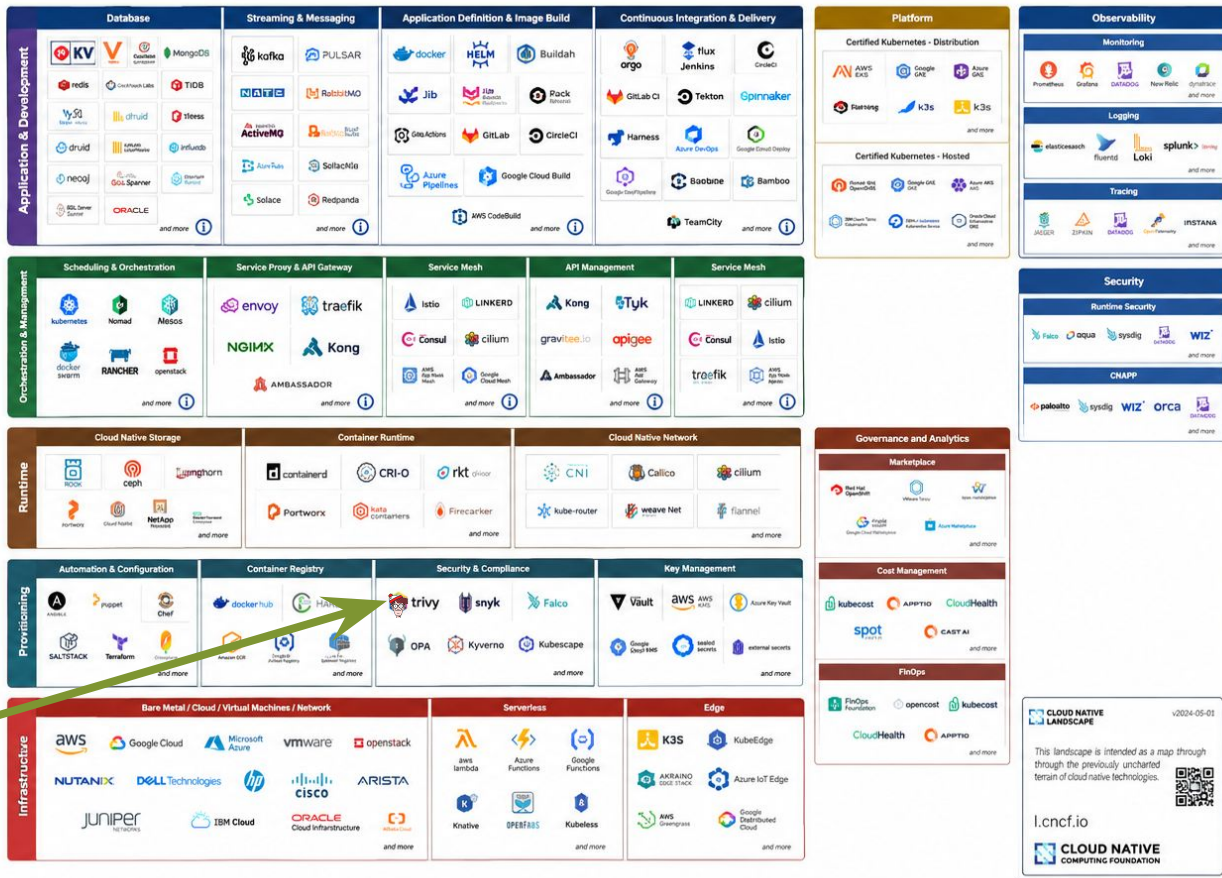


# The Cloud-Native Grocery Store

Where's Waldo?

or Wally?

or Charly?

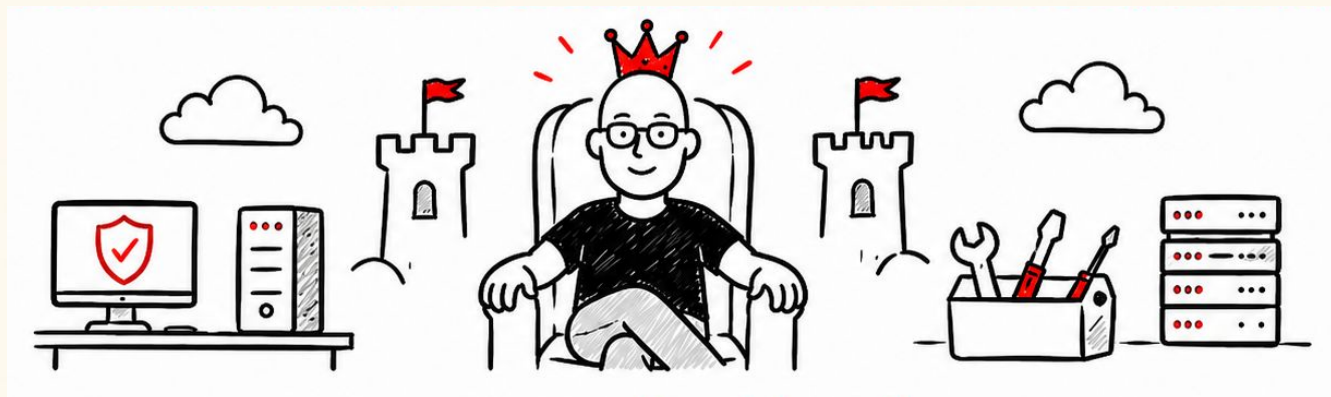




# Where did I start

*(my infra journey)*

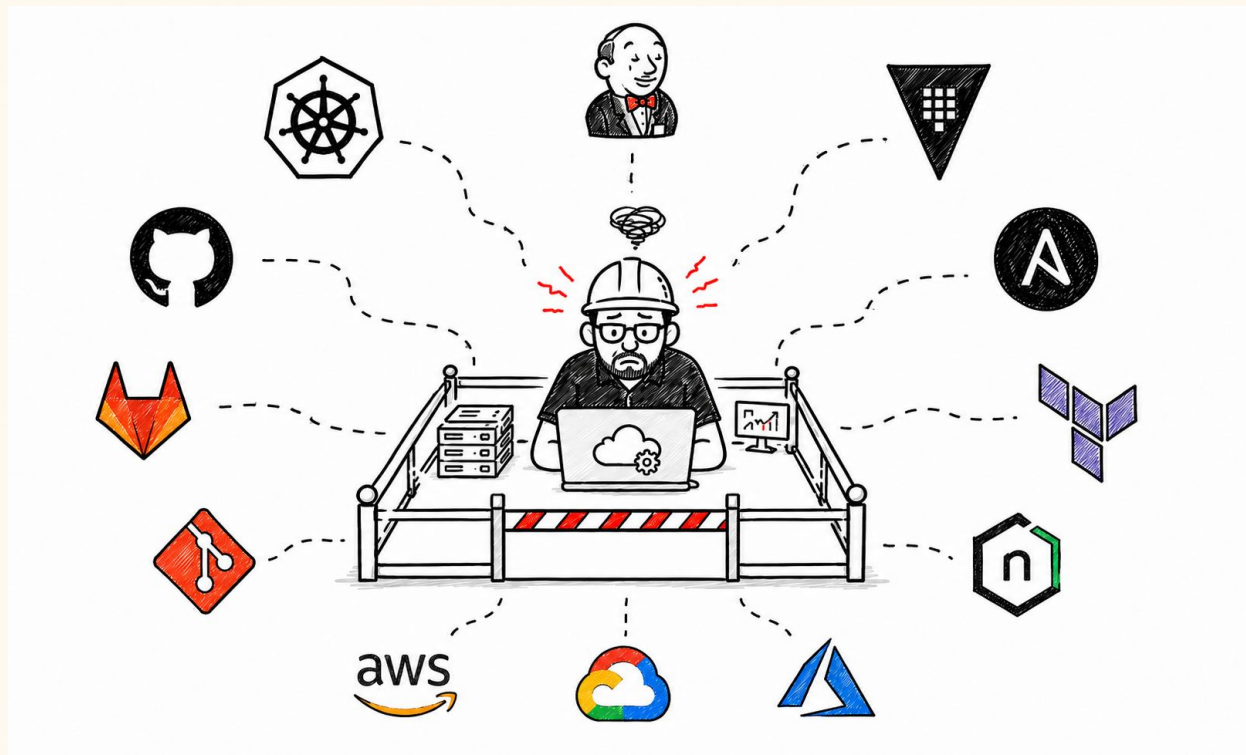






And then I moved...





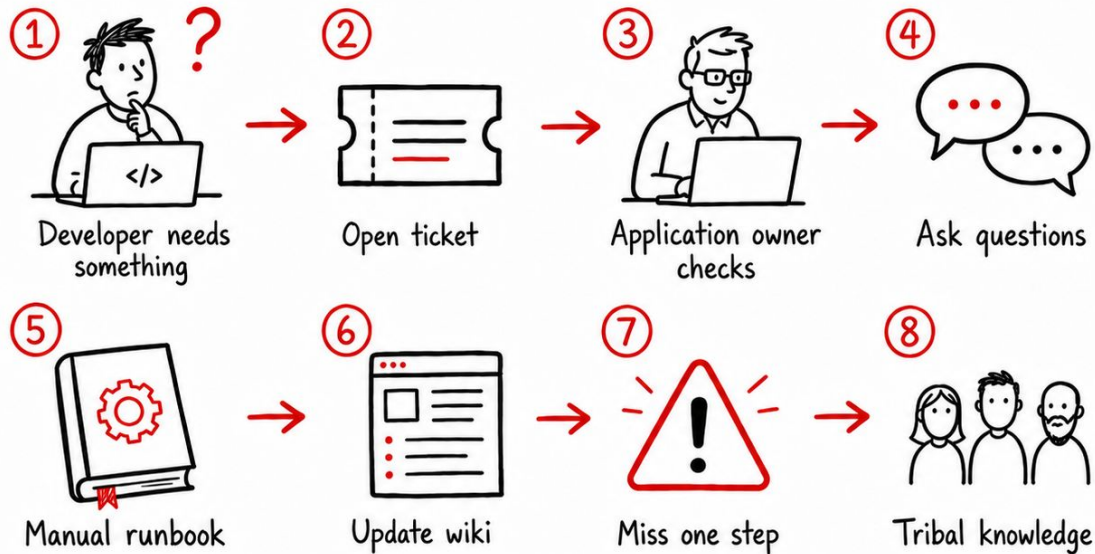


Why are documentation and tickets not enough?



# The classic model

How manual steps turn into tribal knowledge



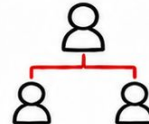
Delays



Manual work

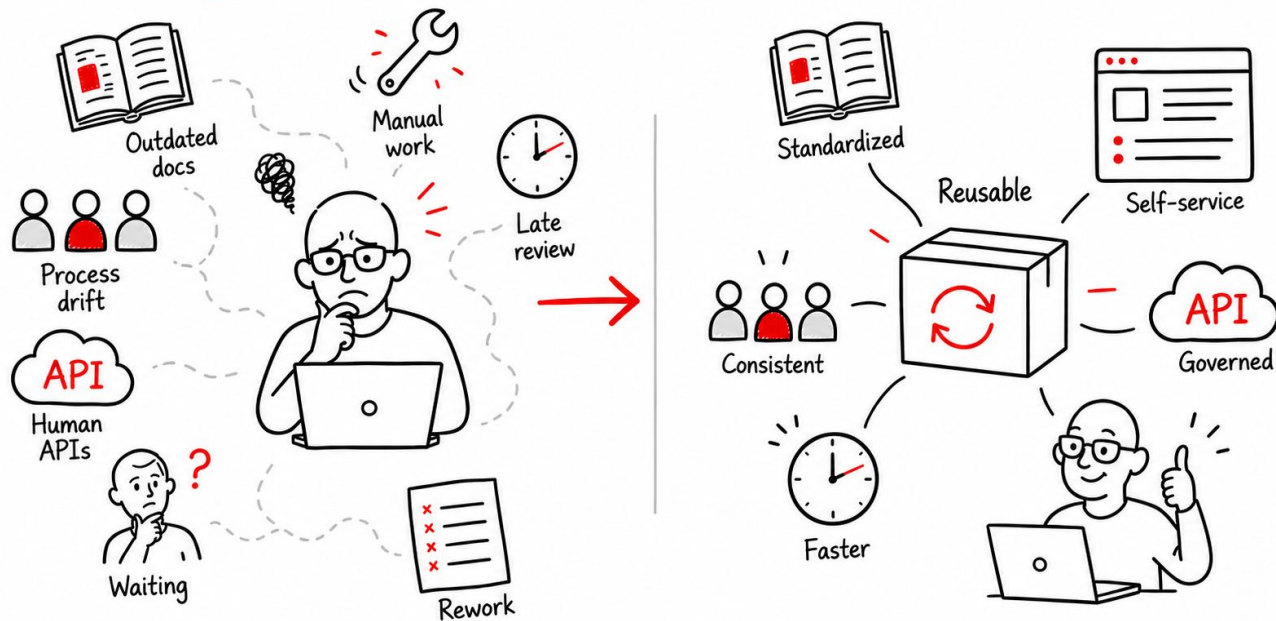


Inconsistency



Dependency on people

## From Manual Chaos to a Reusable Product



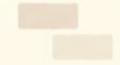


# Platform engineering

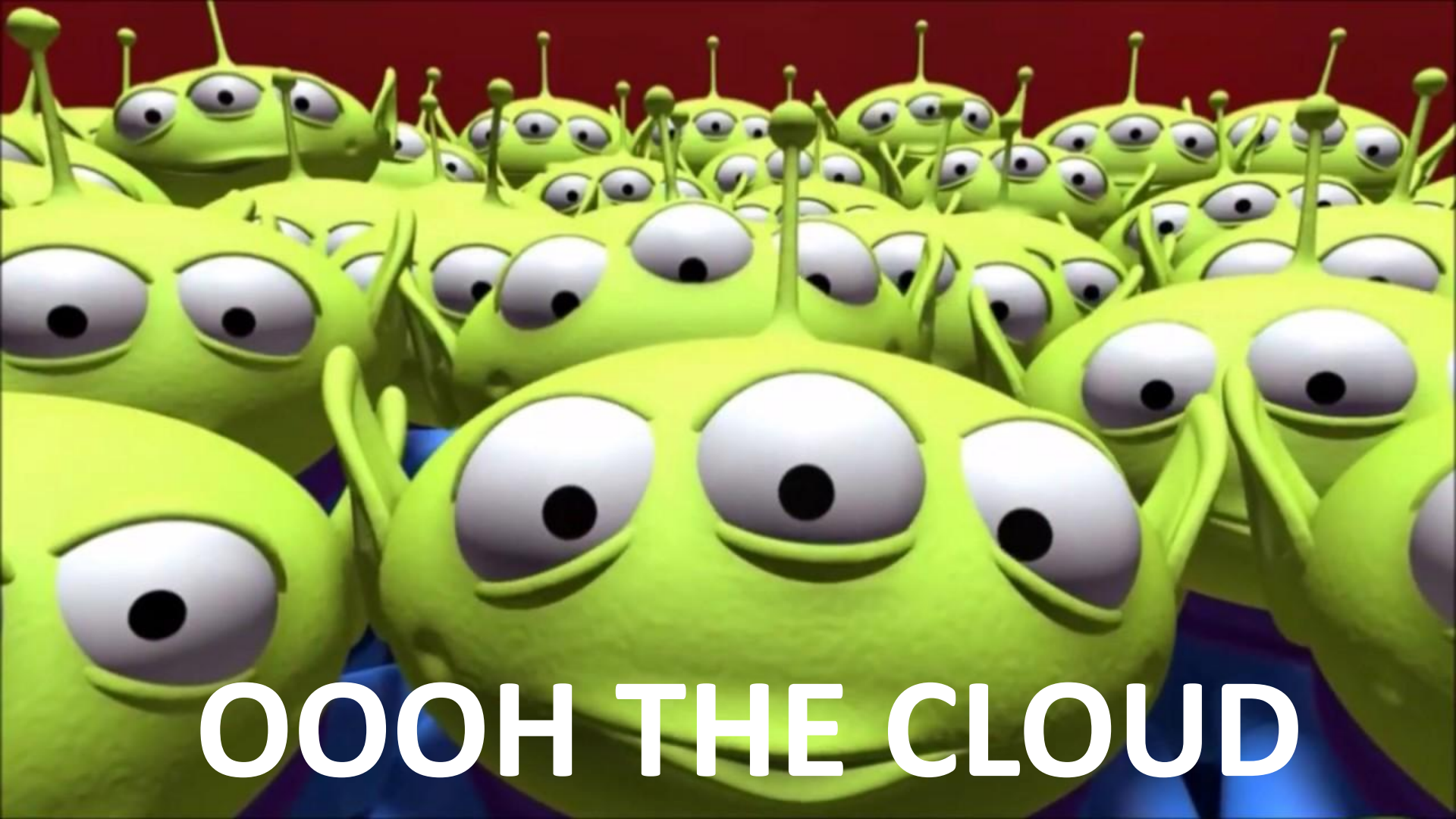




In the past...  
... developers were happy







OOOH THE CLOUD



$$\begin{aligned} H_0 &= H_1 - H_2 - H_3 - \dots \\ H_1 &= \sum_{i=1}^{\infty} \langle \psi_i | \rho | \psi_i \rangle \\ H_2 &= \sum_{i=1}^{\infty} \langle \psi_i | \rho | \psi_i \rangle \\ &\cdot \sum_{j=1}^{\infty} \langle \psi_j | \rho | \psi_j \rangle \\ H_3 &= \rho \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} \sum_{k=1}^{\infty} \langle \psi_i | \rho | \psi_i \rangle \langle \psi_j | \rho | \psi_j \rangle \langle \psi_k | \rho | \psi_k \rangle \\ &= \rho \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} \langle \psi_i | \rho | \psi_i \rangle \langle \psi_j | \rho | \psi_j \rangle \end{aligned}$$

$$G_7 - P_7 LS$$

YOU BUILD IT



YOU RUN ~~AWAY~~ IT

Shift left:  
Increasing  
cognitive load



Dev  
Ops

DevOps

FinOps

MLOps

GitOps

DataOps

DevSecOps

BizDevOps

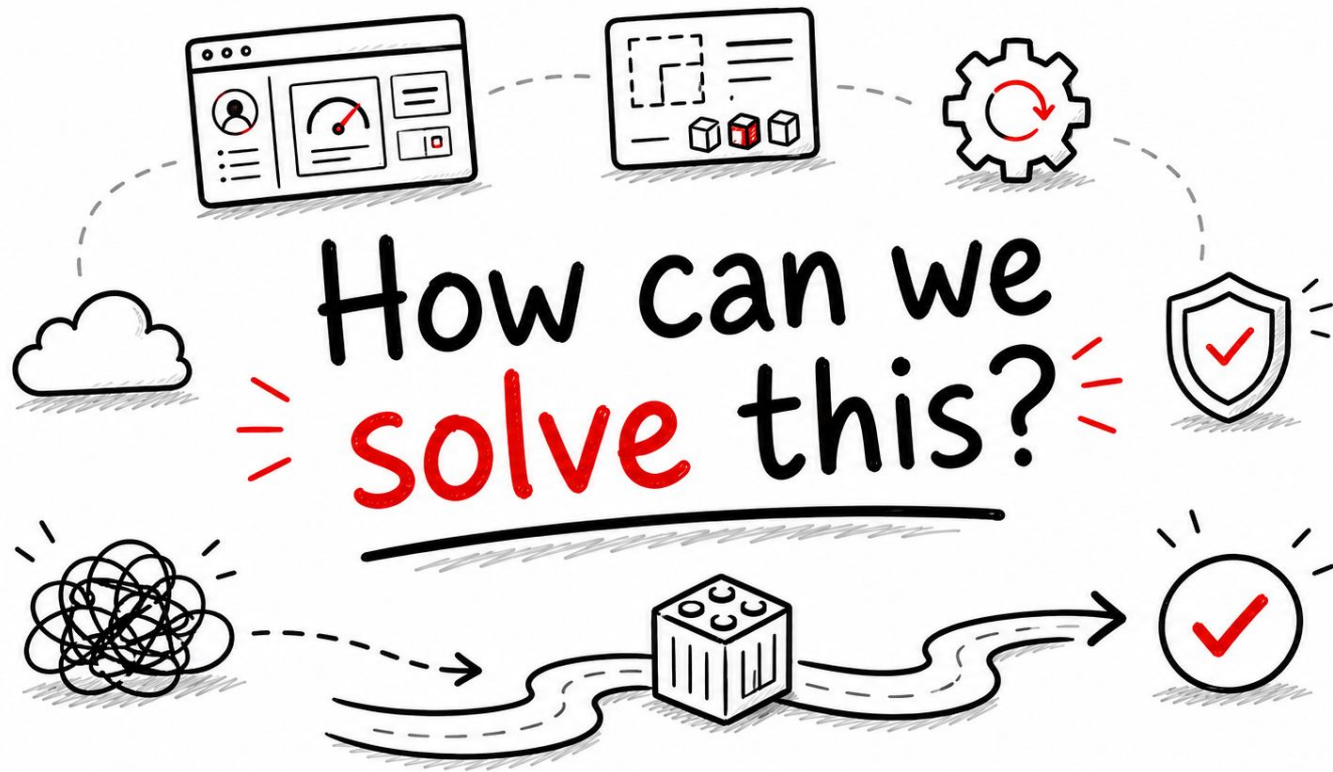
BizDevSecOps

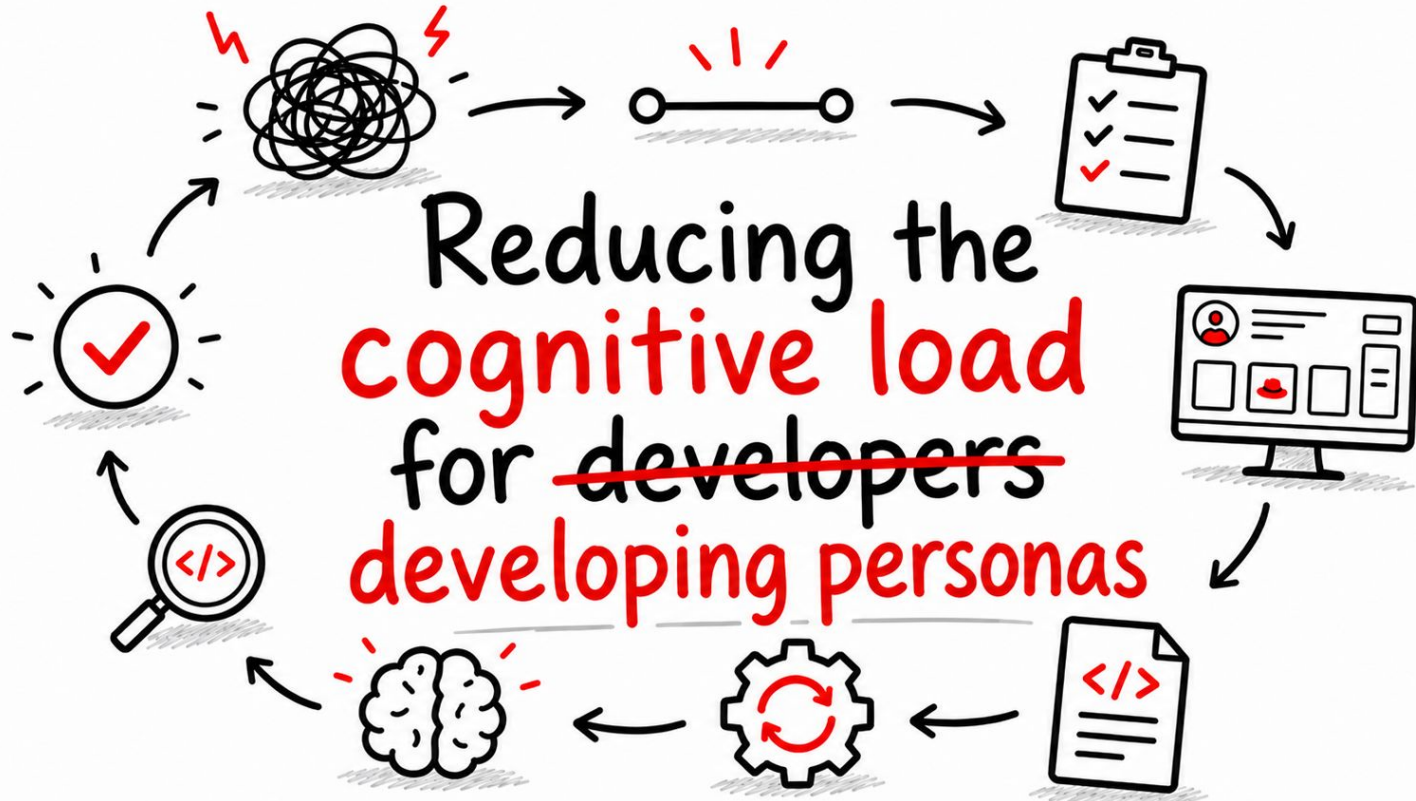
BizDevSecFinMLOps

*It's not just  
app development  
anymore*

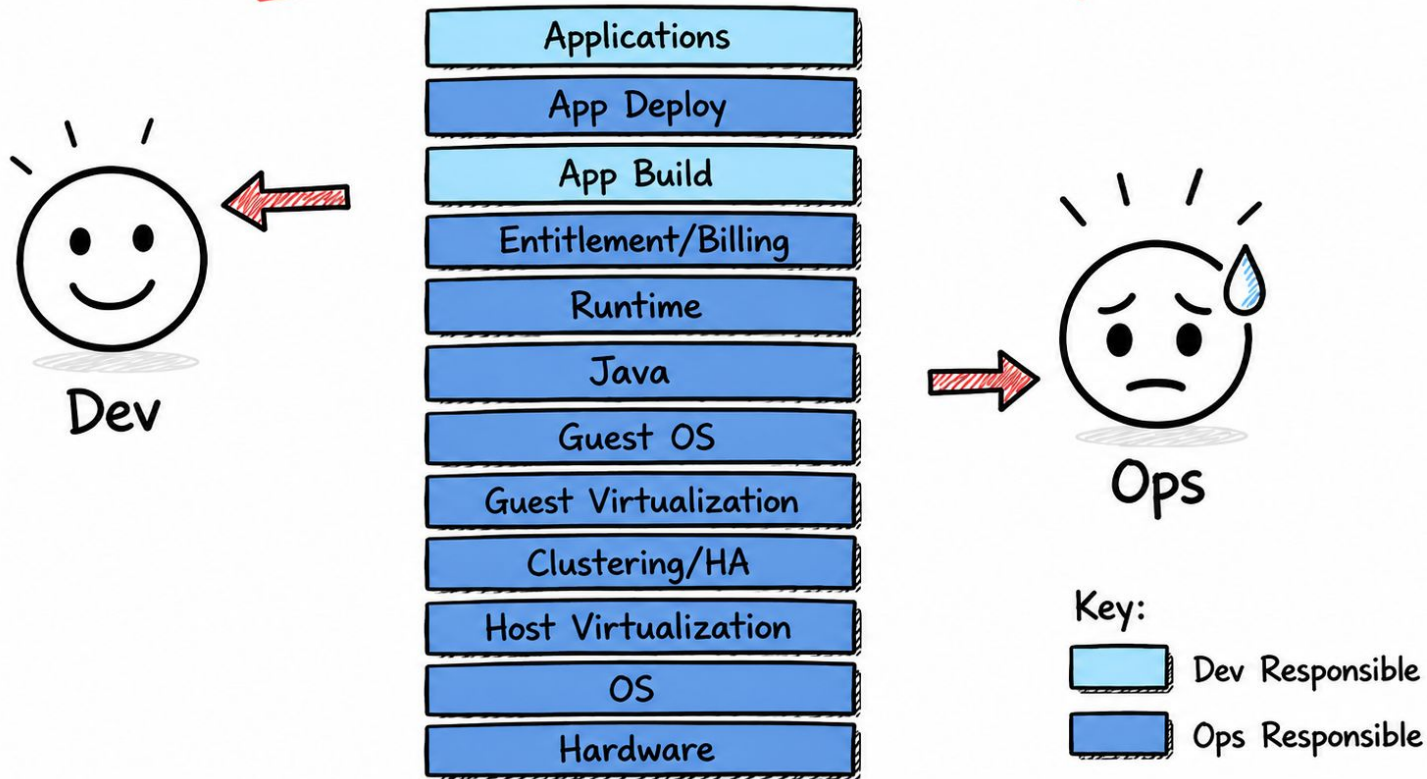






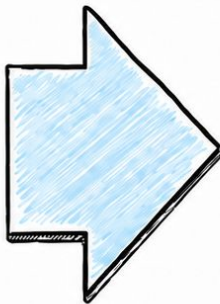
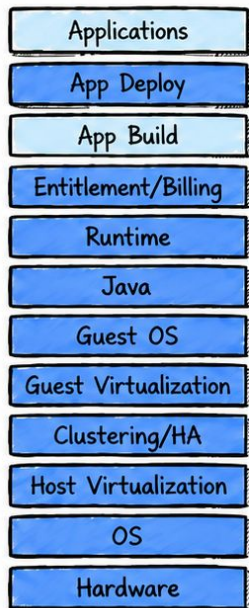


# Traditional App Delivery

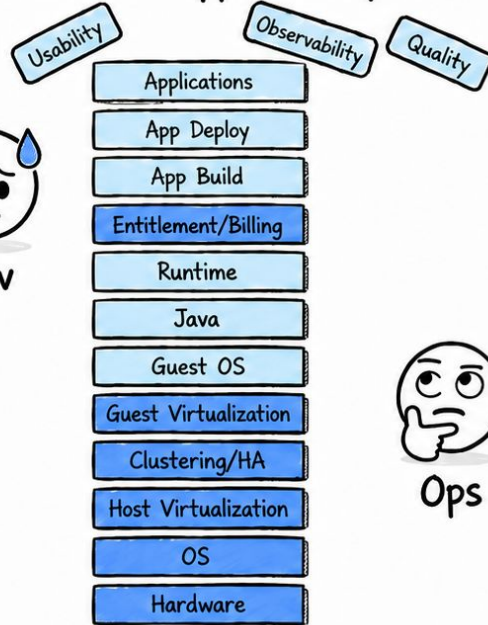


# Platform engineering

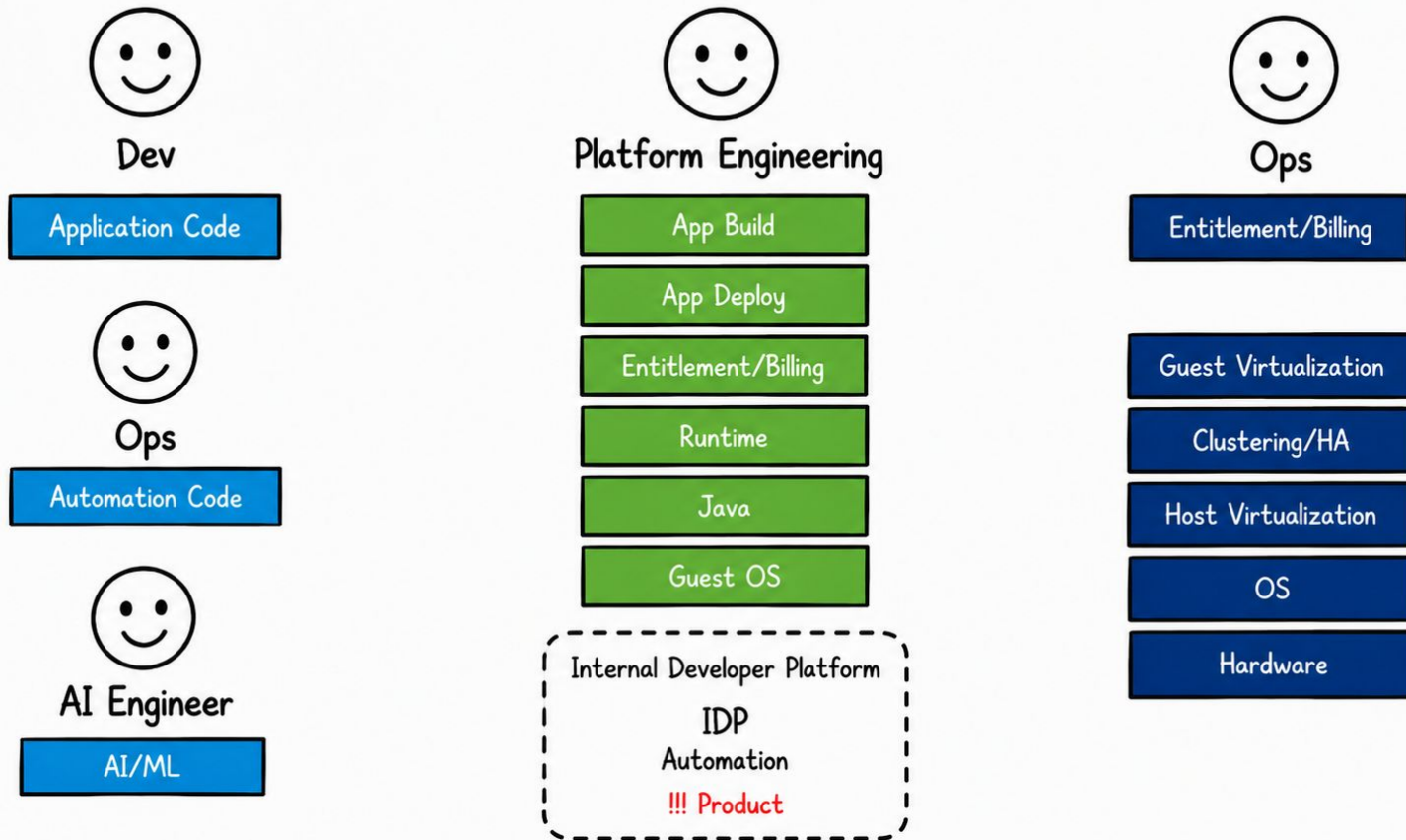
## Traditional App Delivery



## Modern App Delivery



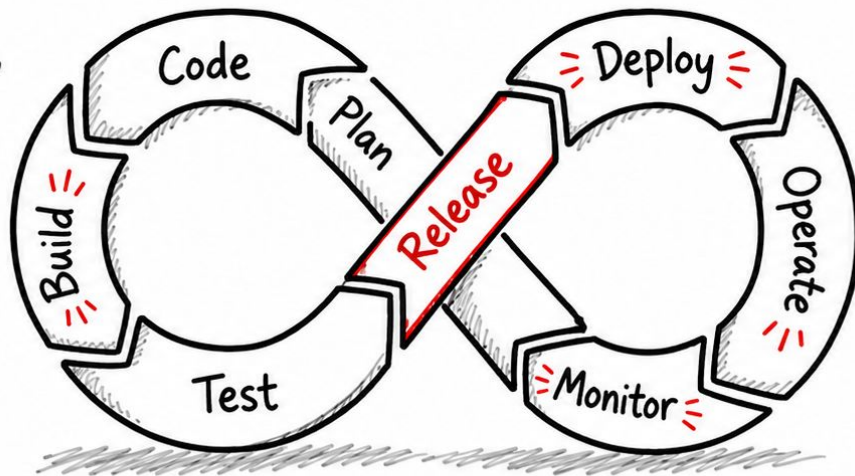
# = Platform engineering =



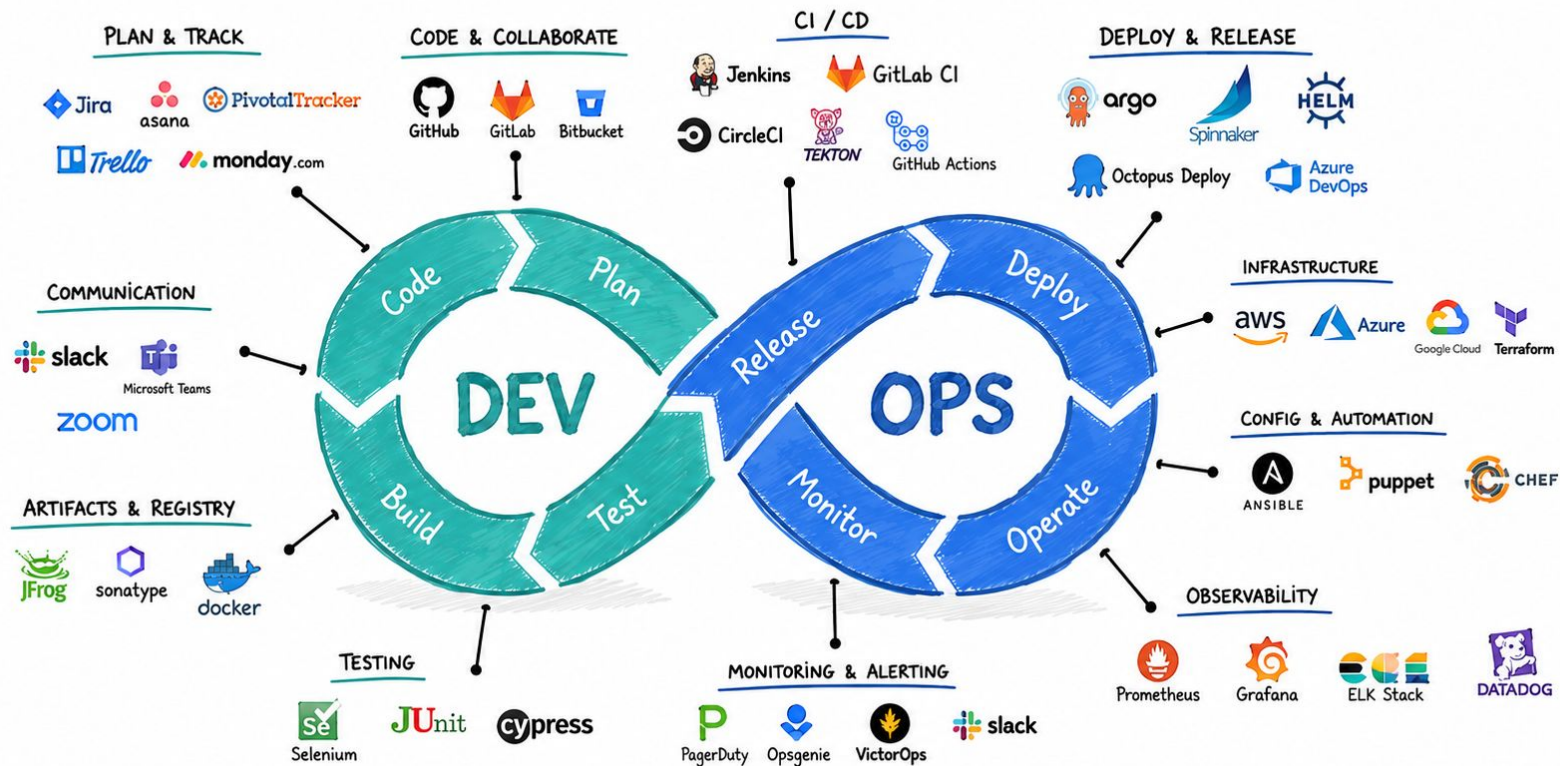
# = The platform engineering recipe =

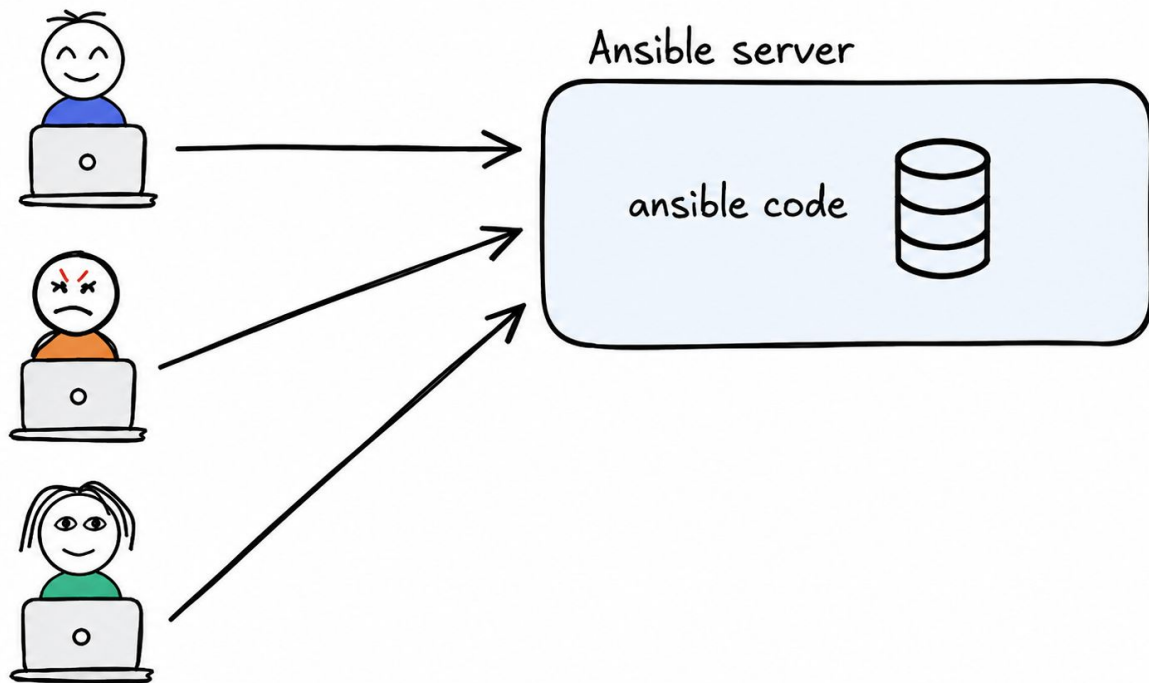
## Ingredients

- Integrated Development Environments
- Version Control Systems
- Resources - containers, databases, storage, compute, transit, etc.
- Runtimes and frameworks
- Artefact, container registry
- Kubernetes
- Observability
- Vulnerability scanners
- CI tools
- GitOps/CD tools
- Infrastructure as Code tools

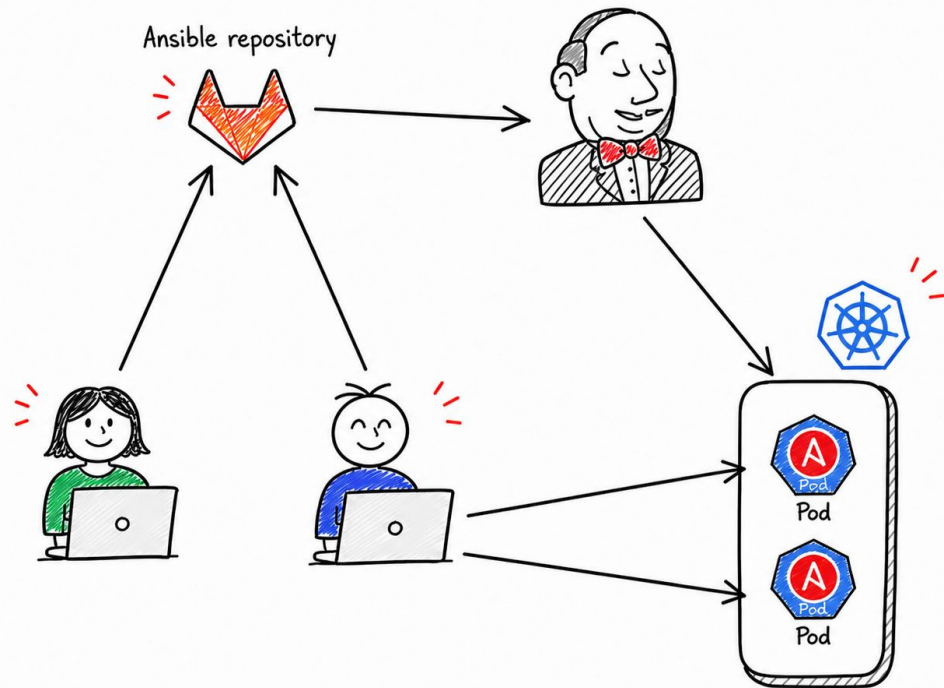


# "Bake your own platform"

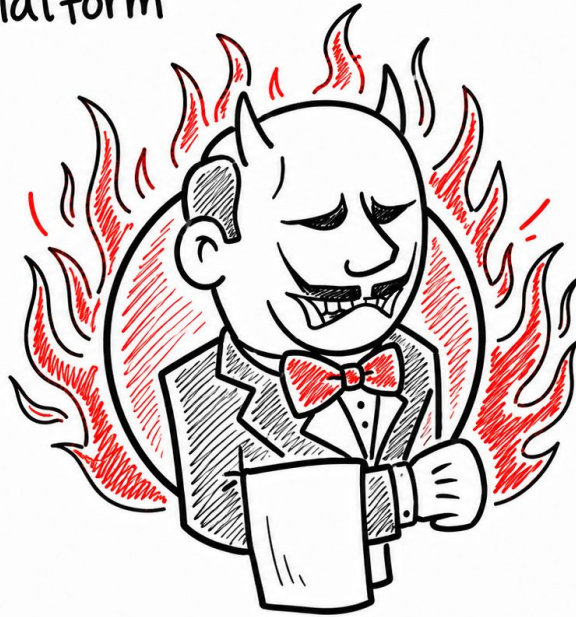




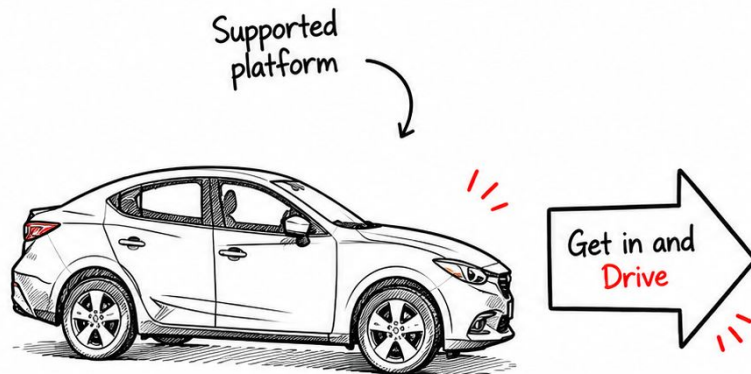
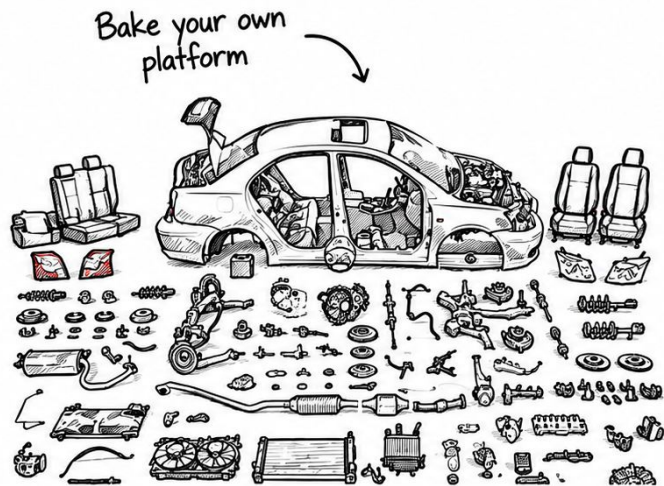
# Bake your own platform



Bake your own platform

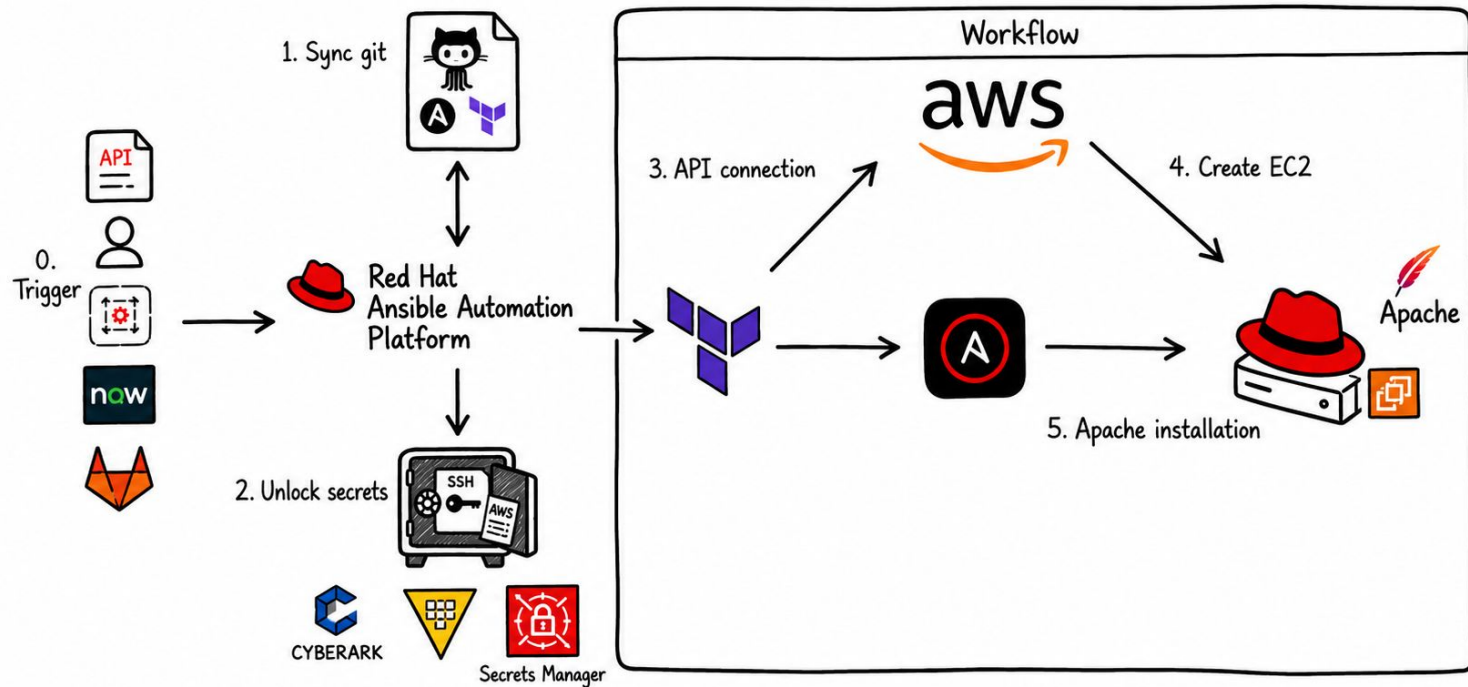


# ⚡ Bake your own platform vs supported platform ⚡



# Ansible Automation Platform

Infra as Code + Config as Code in a unique workflow



# Log in to your account

Enter your credentials.

Username \*

Password \*

Log in



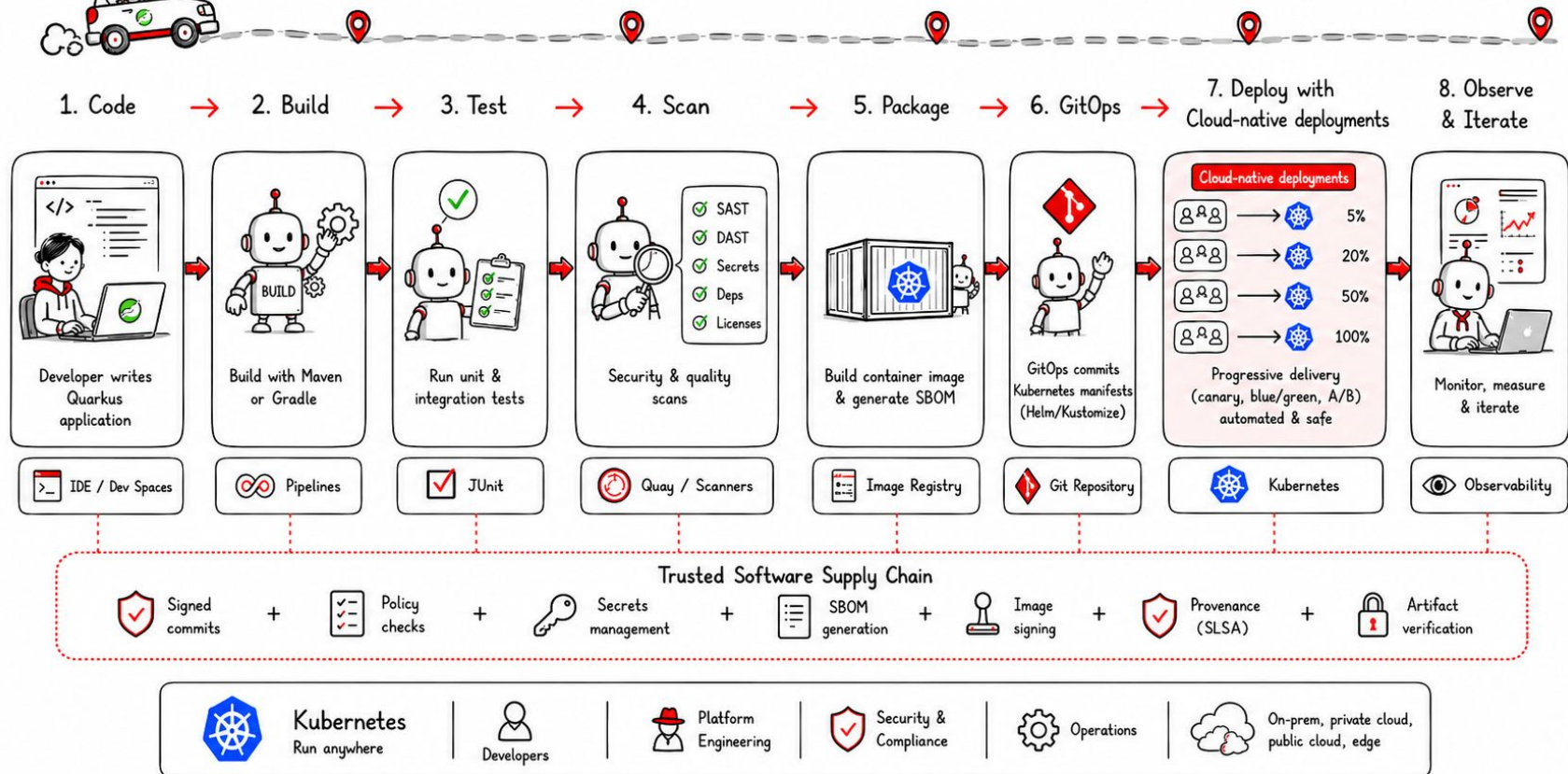
**Red Hat**  
Ansible Automation Platform



Well... I faced the  
Need for a platform as well.

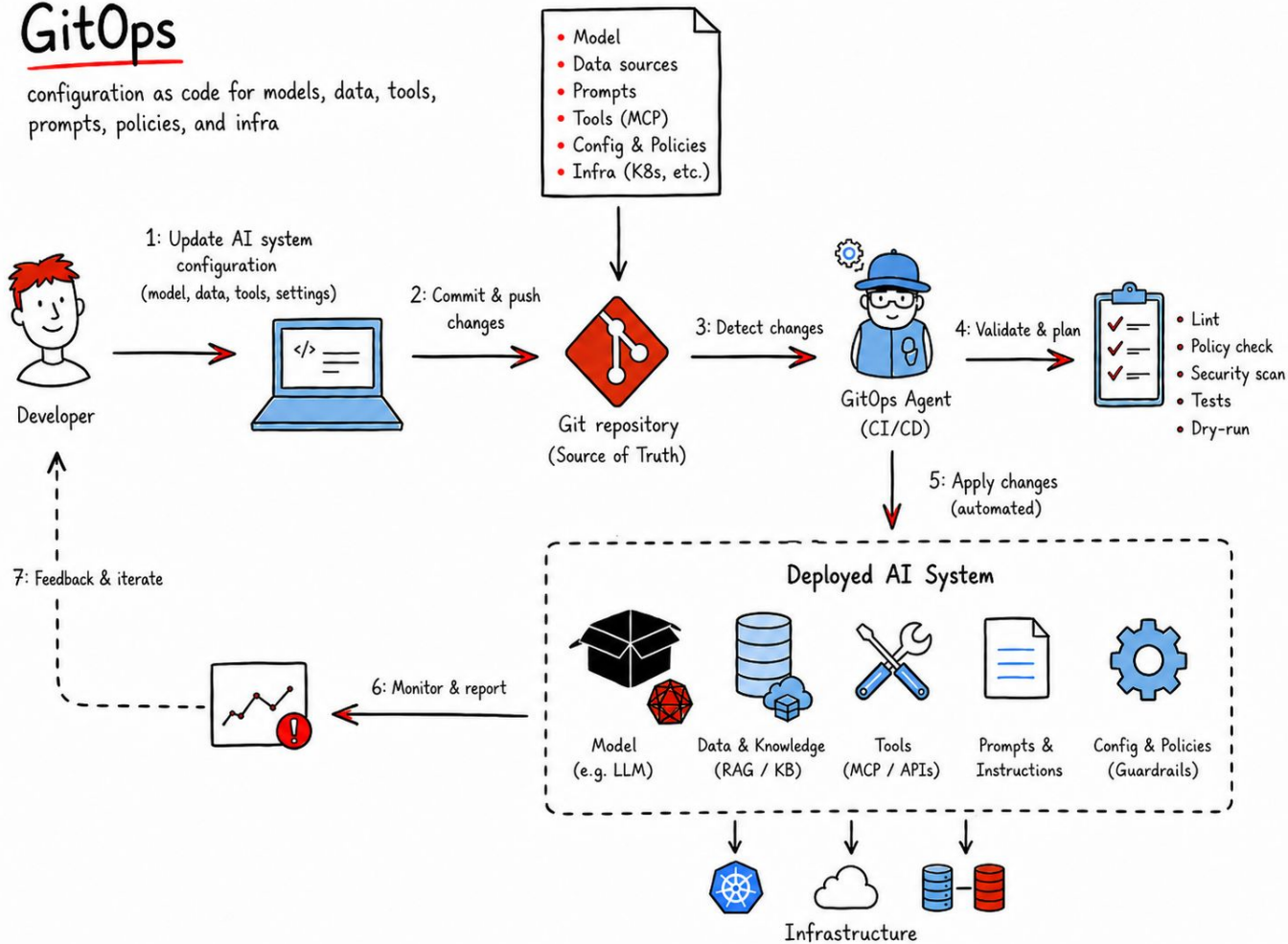


# The road to deploy a ~~Spring Boot~~ Quarkus application on Kubernetes



# GitOps

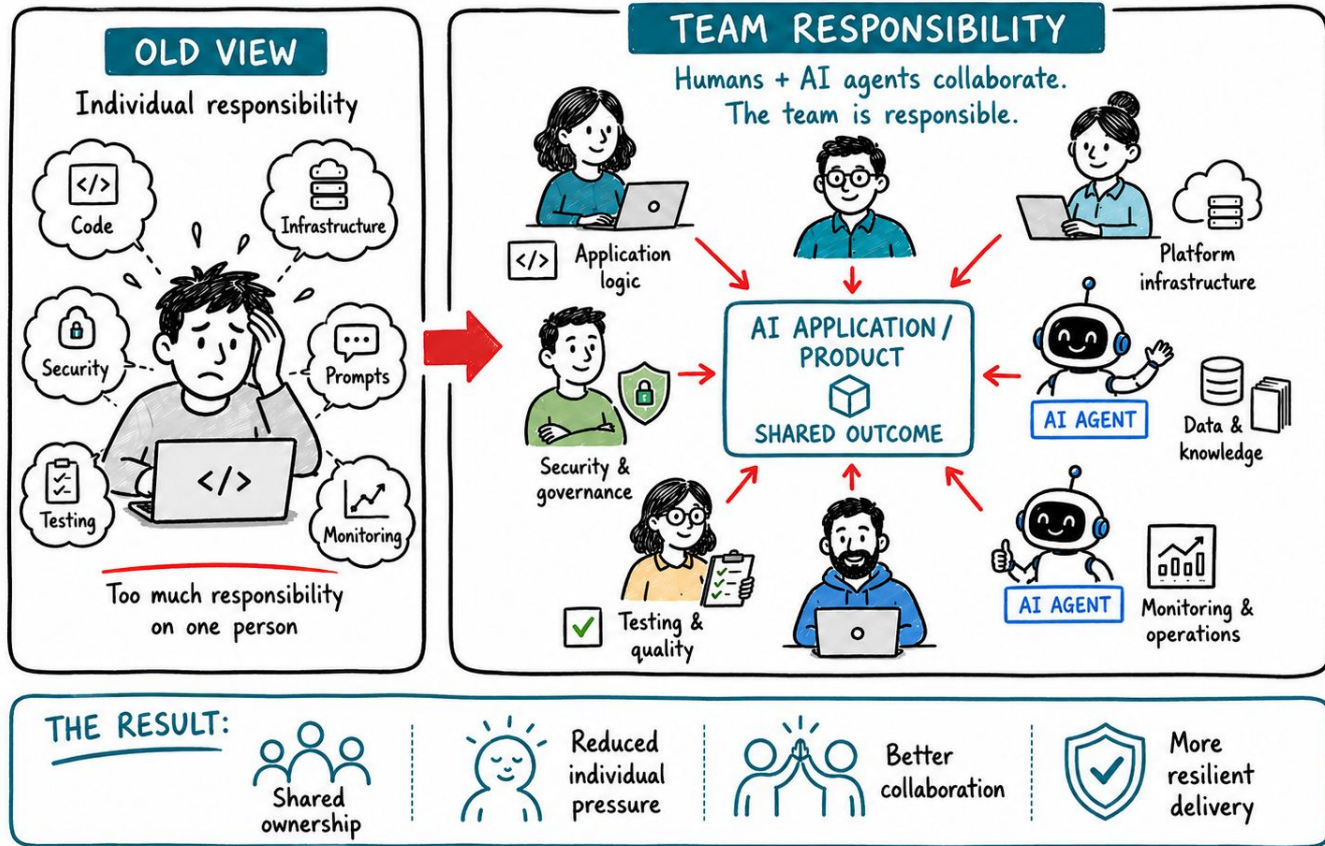
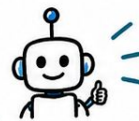
configuration as code for models, data, tools, prompts, policies, and infra





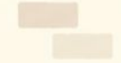
# AI delivery is a team initiative

Responsibility shifts from one individual to a human-and-agent team.

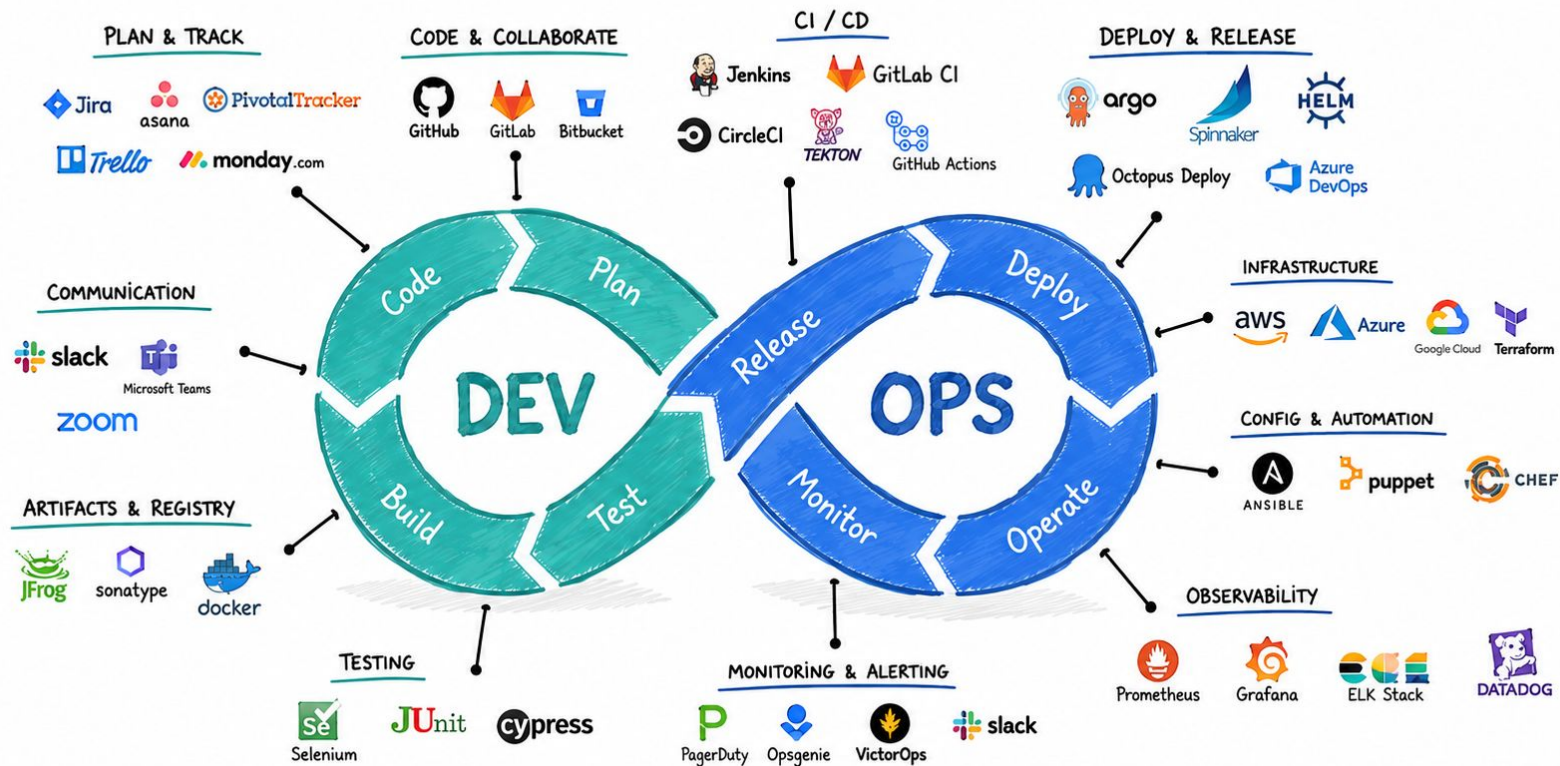




But.... A platform alone isn't  
fixing everything!



# "Bake your own platform"







### Developers

Write code (Java, etc.)



# Backstage

Common front door for Infra & Developers



### Infra (SRE/Platform)

Write automation code  
(Ansible, Terraform, etc.)

#### Self-Service



DB  
Request



Kafka  
Topic Request



Network  
Configs



Firewall  
Changes

...

#### Developer Platform



Kubernetes



CI/CD



Databases



Kafka

...

#### Infra Automation Platform



Ansible  
Automation



Terraform  
IaC

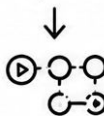


Automation  
Jobs

...



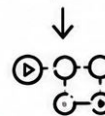
Git repositories



CI/CD pipelines

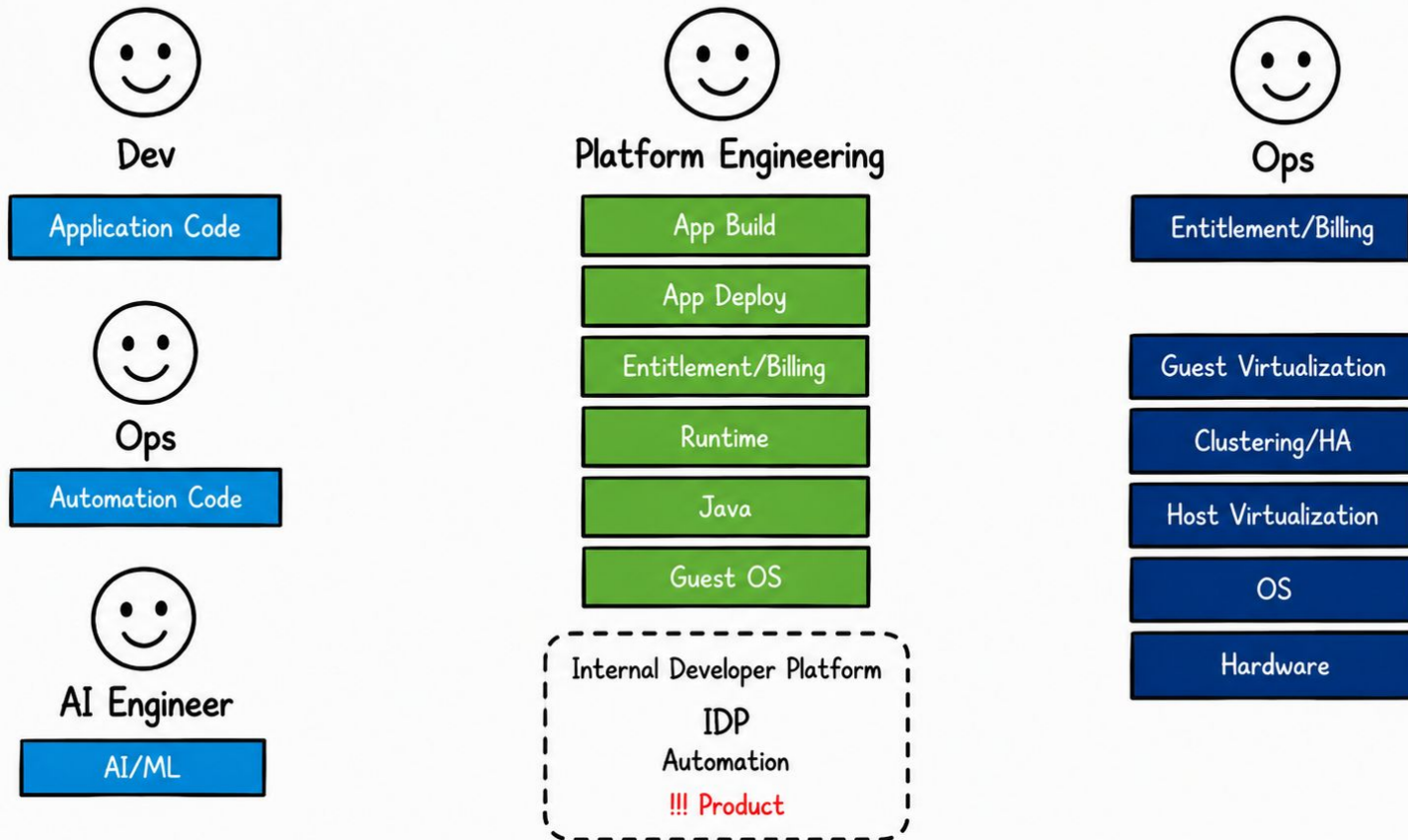


Git repositories



CI/CD pipelines

# = Platform engineering =





# Thanks

## *Questions?*

***maarten-vandeperre.be***

